

Linear Algebra Mastery

Lesson 2: Linear Independence and Basis

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1 Introduction

A set of n -vectors $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n$ are linearly independent if the only solution to $c_1\vec{v}_1 + c_2\vec{v}_2 + \dots + c_n\vec{v}_n$ is zero, and $c_1 = c_2 = \dots = c_n = 0$ where $c_i \in \mathbb{R}$. If any $c_i \neq 0$, then the vectors are *linearly dependent*. In machine learning, dependent vectors carry redundant information. This causes multicollinearity, rendering the model weights unstable.

A basis is the minimum set of vectors that are linearly independent and span the entire space, meaning the vectors can reach every point in space. The number of basis vectors are equal to the TRUE *dimensionality* of a dataset.

2 Formal Definitions

2.1 Linear Independence

Linear independence holds if:

$$c_1\vec{v}_1 + c_2\vec{v}_2 + \dots + c_n\vec{v}_n = \vec{0}$$

where;

$$c_1 = c_2 = c_3 = \dots = c_n = 0$$

If $c_i \neq 0$, then the set of vectors is said to be linearly dependent.

Possible Outcomes

1. Linear independence - Each vector add a genuine new information
2. Linear dependence - Redundant vector(s) present, causes multicollinearity.

2.2 Basis

For a set of vectors to be a basis vector the following two properties must hold:

1. Vectors must be linearly independent
2. Span the entire space. The vectors must reach every point in that space

2.3 Standard Basis

There are standard basis in every space, e.g.

$$\text{Standard basis in } \mathbb{R}^2 \rightarrow \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\}$$

$$\text{Standard basis in } \mathbb{R}^3 \rightarrow \left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\}$$

$$\text{Standard basis in } \mathbb{R}^n \rightarrow \left\{ \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix} \right\}$$

3 Proof: A Set Containing the Zero Vector is Always Linearly Dependent

We want to prove that a set of n -vectors $S = \{\vec{0}, \vec{v}_1, \vec{v}_2, \vec{v}_3, \dots, \vec{v}_n\}$ is linearly dependent because it contains the zero vector. For a linearly independent set of vectors, the linear combination is $c_0\vec{v}_0 + c_1\vec{v}_1 + c_2\vec{v}_2 + \dots + c_n\vec{v}_n = \vec{0}$.

Let $c_0 = 1$, $v_i \in S$ and $c_i \in \mathbb{R}$

Then;

$$1 \cdot \vec{0} + 0 \cdot \vec{v}_1 + 0 \cdot \vec{v}_2 + \dots + 0 \cdot \vec{v}_n = \vec{0}$$

The above equation shows that there exist c_i 's that are not all zeros. In conclusion, $\{\vec{0}, \vec{v}_1, \dots, \vec{v}_n\}$ is a linearly dependent set.

4 Python Results

4.1 Linear Independence Check

Vector $v_1 = (1, 2)$ and $v_2 = (3, 5)$. The rank for the two stacked vectors is 2, meaning that every point in $\mathbb{R}^2$ is reachable, hence linearly independent.

4.2 Linear Dependence Check

Vectors $v_3 = (2, 4)$ and $v_4 = (1, 2)$ are linearly dependent with rank = 1. This means that at least one of the vectors is redundant.

4.3 Standard Basis Verification

The rank for standard basis in $\mathbb{R}^2$ is 2, while in $\mathbb{R}^3$ is 3. This means that with standard basis vectors, you can reach every point in a vector space.

4.4 ML Connection — Feature Matrix

The feature matrix $X = \begin{pmatrix} 1 & 2 \\ 3 & 5 \\ 2 & 4 \end{pmatrix}$ have a rank = 2 when checking among features. With 2 features present, it means that the columns are not redundant. Always know what you are checking before interpreting rank, because columns and rows have different ranks. In our case, for example, At least one row is redundant.

5 ML Connection

Before training a model, check for redundant features. Redundant features add no new information causing multicollinearity, which makes a model weights to be unstable.

6 Reflection

The hardest concept today was to proof that a set of vectors containing the zero vector is always dependent. I was surprised by how you can check redundancy for features or rows.